

class10: Halloween Candy Miniproject

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```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.3.3

```
library(skimr)
library(ggplot2)
library(ggrepel)
```

Warning: package 'ggrepel' was built under R version 4.3.3

```
library(corrplot)
```

Warning: package 'corrplot' was built under R version 4.3.3

```
candy_file <- "candy-data.csv"

candy <- read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisp	rice	wafer
100 Grand	1	0	1		0	0			1
3 Musketeers	1	0	0		0	1			0
One dime	0	0	0		0	0			0
One quarter	0	0	0		0	0			0
Air Heads	0	1	0		0	0			0
Almond Joy	1	0	0		1	0			0
	hard bar	pluribus	sugar	percent	price	percent	win	percent	
100 Grand	0	1	0	0.732	0.860	66.97	173		
3 Musketeers	0	1	0	0.604	0.511	67.60	294		

One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

There are 85 different candies in this dataset.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in the dataset.

Q3. What is your favorite candy in the dataset and what is its winpercent value?

```
candy["Reese's Peanut Butter cup",]$winpercent
```

```
[1] 84.18029
```

My favorite candy is Reese's peanut butter cups, which has a 84.18 win percent.

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat",]$winpercent
```

```
[1] 76.7686
```

Kitkat has a 76.76% win percent.

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars",]$winpercent
```

```
[1] 49.6535
```

Tootsie Roll Snack Bars have a 49.65% win percent.

```
candy %>% select(winpercent)
```

	winpercent
100 Grand	66.97173
3 Musketeers	67.60294
One dime	32.26109
One quarter	46.11650
Air Heads	52.34146
Almond Joy	50.34755
Baby Ruth	56.91455
Boston Baked Beans	23.41782
Candy Corn	38.01096
Caramel Apple Pops	34.51768
Charleston Chew	38.97504
Chewey Lemonhead Fruit Mix	36.01763
Chiclets	24.52499
Dots	42.27208
Dum Dums	39.46056
Fruit Chews	43.08892
Fun Dip	39.18550
Gobstopper	46.78335
Haribo Gold Bears	57.11974
Haribo Happy Cola	34.15896
Haribo Sour Bears	51.41243
Haribo Twin Snakes	42.17877
Hershey's Kisses	55.37545
Hershey's Krackel	62.28448
Hershey's Milk Chocolate	56.49050
Hershey's Special Dark	59.23612
Jawbusters	28.12744
Junior Mints	57.21925
Kit Kat	76.76860
Laffy Taffy	41.38956
Lemonhead	39.14106
Lifesavers big ring gummies	52.91139

Peanut butter M&M's	71.46505
M&M's	66.57458
Mike & Ike	46.41172
Milk Duds	55.06407
Milky Way	73.09956
Milky Way Midnight	60.80070
Milky Way Simply Caramel	64.35334
Mounds	47.82975
Mr Good Bar	54.52645
Nerds	55.35405
Nestle Butterfinger	70.73564
Nestle Crunch	66.47068
Nik L Nip	22.44534
Now & Later	39.44680
Payday	46.29660
Peanut M&Ms	69.48379
Pixie Sticks	37.72234
Pop Rocks	41.26551
Red vines	37.34852
Reese's Miniatures	81.86626
Reese's Peanut Butter cup	84.18029
Reese's pieces	73.43499
Reese's stuffed with pieces	72.88790
Ring pop	35.29076
Rolo	65.71629
Root Beer Barrels	29.70369
Runts	42.84914
Sixlets	34.72200
Skittles original	63.08514
Skittles wildberry	55.10370
Nestle Smarties	37.88719
Smarties candy	45.99583
Snickers	76.67378
Snickers Crisper	59.52925
Sour Patch Kids	59.86400
Sour Patch Tricksters	52.82595
Starburst	67.03763
Strawberry bon bons	34.57899
Sugar Babies	33.43755
Sugar Daddy	32.23100
Super Bubble	27.30386
Swedish Fish	54.86111
Tootsie Pop	48.98265

Tootsie Roll Juniors	43.06890
Tootsie Roll Midgies	45.73675
Tootsie Roll Snack Bars	49.65350
Trolli Sour Bites	47.17323
Twix	81.64291
Twizzlers	45.46628
Warheads	39.01190
Welch's Fruit Snacks	44.37552
Werther's Original Caramel	41.90431
Whoppers	49.52411

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

```
skim(candy)
```

```
Error in options$skimr_include_summary %||% TRUE: could not find function "%||%"
```

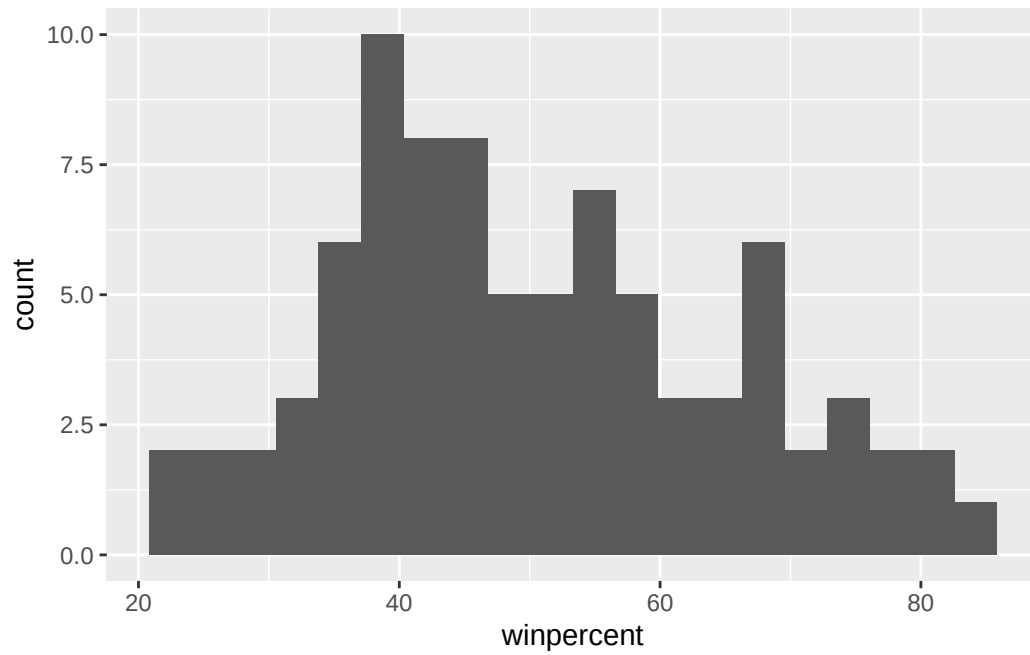
The winpercent variable looks like it is on a different scale to the majority of the dataset.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

A 1 represents the specific candy is chocolate based and a 0 means it is not.

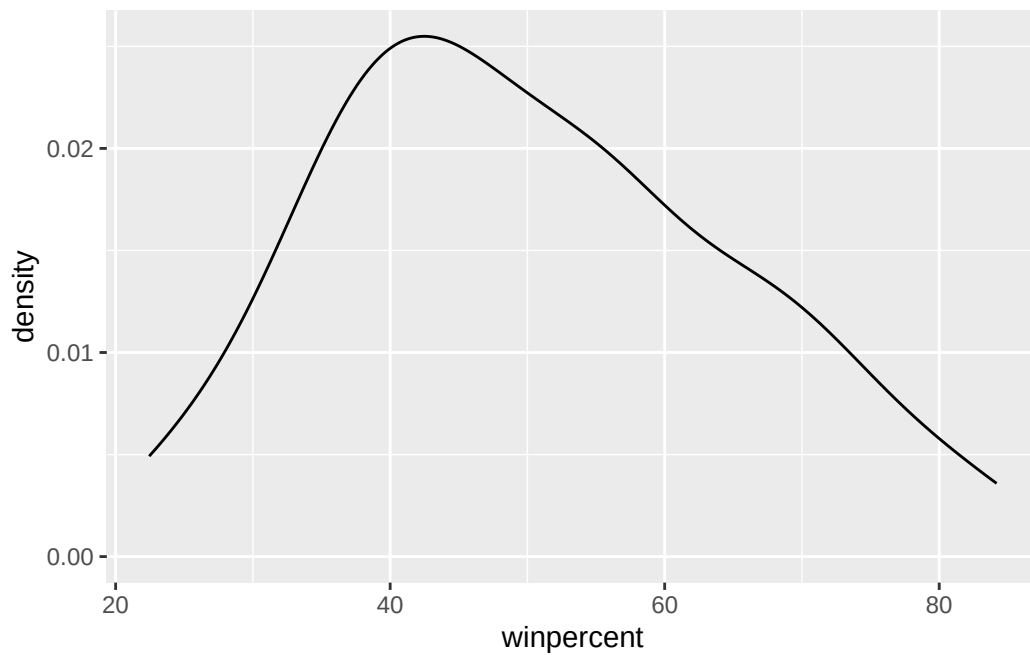
Q8. Plot a histogram of winpercent values

```
ggplot(candy) +aes(winpercent) +geom_histogram(bins = 20)
```



Q9. Is the distribution of winpercent values symmetrical?

```
ggplot(candy) + aes(winpercent) +geom_density()
```



The distribution is not symmetric.

Q10. Is the center of the distribution above or below 50%?

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

The mean is 50.32 which is above 50% but the median is 47.83 which is below 50%. We trust median more as a measure of centrality when there are outliers.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# 1. Find all chocolate candy in the dataset
```

```
choc.inds <- as.logical(candy$chocolate)
choc.inds
```

```
[1] TRUE TRUE FALSE FALSE FALSE TRUE TRUE FALSE FALSE FALSE TRUE FALSE
[13] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE
[25] TRUE TRUE FALSE TRUE TRUE FALSE FALSE FALSE TRUE TRUE FALSE TRUE
[37] TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE FALSE FALSE TRUE
[49] FALSE FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE FALSE FALSE TRUE
[61] FALSE FALSE TRUE FALSE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE
[73] FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE FALSE FALSE FALSE FALSE
[85] TRUE
```

```
choc.candy <- candy[choc.inds,]
choc.candy
```

	chocolate	fruity	caramel	peanut	almond	nougat
100 Grand	1	0	1		0	0
3 Musketeers	1	0	0		0	1
Almond Joy	1	0	0		1	0
Baby Ruth	1	0	1		1	1
Charleston Chew	1	0	0		0	1
Hershey's Kisses	1	0	0		0	0
Hershey's Krackel	1	0	0		0	0
Hershey's Milk Chocolate	1	0	0		0	0
Hershey's Special Dark	1	0	0		0	0

Junior Mints	1	0	0	0	0
Kit Kat	1	0	0	0	0
Peanut butter M&M's	1	0	0	1	0
M&M's	1	0	0	0	0
Milk Duds	1	0	1	0	0
Milky Way	1	0	1	0	1
Milky Way Midnight	1	0	1	0	1
Milky Way Simply Caramel	1	0	1	0	0
Mounds	1	0	0	0	0
Mr Good Bar	1	0	0	1	0
Nestle Butterfinger	1	0	0	1	0
Nestle Crunch	1	0	0	0	0
Peanut M&Ms	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0
Reese's pieces	1	0	0	1	0
Reese's stuffed with pieces	1	0	0	1	0
Rolo	1	0	1	0	0
Sixlets	1	0	0	0	0
Nestle Smarties	1	0	0	0	0
Snickers	1	0	1	1	1
Snickers Crisper	1	0	1	1	0
Tootsie Pop	1	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Tootsie Roll Snack Bars	1	0	0	0	0
Twix	1	0	1	0	0
Whoppers	1	0	0	0	0
	crisped	ricewafer	hard	bar	pluribus
100 Grand		1	0	1	0
3 Musketeers		0	0	1	0
Almond Joy		0	0	1	0
Baby Ruth		0	0	1	0
Charleston Chew		0	0	1	0
Hershey's Kisses		0	0	0	1
Hershey's Krackel		1	0	1	0
Hershey's Milk Chocolate		0	0	1	0
Hershey's Special Dark		0	0	1	0
Junior Mints		0	0	0	1
Kit Kat		1	0	1	0
Peanut butter M&M's		0	0	0	1
M&M's		0	0	0	1
Milk Duds		0	0	0	1
				sugar	percent

Milky Way	0	0	1	0	0.604
Milky Way Midnight	0	0	1	0	0.732
Milky Way Simply Caramel	0	0	1	0	0.965
Mounds	0	0	1	0	0.313
Mr Good Bar	0	0	1	0	0.313
Nestle Butterfinger	0	0	1	0	0.604
Nestle Crunch	1	0	1	0	0.313
Peanut M&Ms	0	0	0	1	0.593
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's pieces	0	0	0	1	0.406
Reese's stuffed with pieces	0	0	0	0	0.988
Rolo	0	0	0	1	0.860
Sixlets	0	0	0	1	0.220
Nestle Smarties	0	0	0	1	0.267
Snickers	0	0	1	0	0.546
Snickers Crisper	1	0	1	0	0.604
Tootsie Pop	0	1	0	0	0.604
Tootsie Roll Juniors	0	0	0	0	0.313
Tootsie Roll Midgies	0	0	0	1	0.174
Tootsie Roll Snack Bars	0	0	1	0	0.465
Twix	1	0	1	0	0.546
Whoppers	1	0	0	1	0.872

pricepercent winpercent

100 Grand	0.860	66.97173
3 Musketeers	0.511	67.60294
Almond Joy	0.767	50.34755
Baby Ruth	0.767	56.91455
Charleston Chew	0.511	38.97504
Hershey's Kisses	0.093	55.37545
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050
Hershey's Special Dark	0.918	59.23612
Junior Mints	0.511	57.21925
Kit Kat	0.511	76.76860
Peanut butter M&M's	0.651	71.46505
M&M's	0.651	66.57458
Milk Duds	0.511	55.06407
Milky Way	0.651	73.09956
Milky Way Midnight	0.441	60.80070
Milky Way Simply Caramel	0.860	64.35334
Mounds	0.860	47.82975
Mr Good Bar	0.918	54.52645

Nestle Butterfinger	0.767	70.73564
Nestle Crunch	0.767	66.47068
Peanut M&Ms	0.651	69.48379
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029
Reese's pieces	0.651	73.43499
Reese's stuffed with pieces	0.651	72.88790
Rolo	0.860	65.71629
Sixlets	0.081	34.72200
Nestle Smarties	0.976	37.88719
Snickers	0.651	76.67378
Snickers Crisper	0.651	59.52925
Tootsie Pop	0.325	48.98265
Tootsie Roll Juniors	0.511	43.06890
Tootsie Roll Midgies	0.011	45.73675
Tootsie Roll Snack Bars	0.325	49.65350
Twix	0.906	81.64291
Whoppers	0.848	49.52411

```
# 2. Extract their `winpercent` values
```

```
choc.win <- choc.candy$winpercent
choc.win
```

```
[1] 66.97173 67.60294 50.34755 56.91455 38.97504 55.37545 62.28448 56.49050
[9] 59.23612 57.21925 76.76860 71.46505 66.57458 55.06407 73.09956 60.80070
[17] 64.35334 47.82975 54.52645 70.73564 66.47068 69.48379 81.86626 84.18029
[25] 73.43499 72.88790 65.71629 34.72200 37.88719 76.67378 59.52925 48.98265
[33] 43.06890 45.73675 49.65350 81.64291 49.52411
```

```
# 3. Find the mean of these values
```

```
choc.mean <- mean(choc.win)
choc.mean
```

```
[1] 60.92153
```

```
# 4-6. Do the same for the fruity candy
```

```
fruit.inds <- as.logical(candy$fruity)
fruit.inds
```

```

[1] FALSE FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE TRUE FALSE TRUE
[13] TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE FALSE
[25] FALSE FALSE TRUE FALSE FALSE TRUE TRUE TRUE FALSE FALSE TRUE FALSE
[37] FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE TRUE TRUE FALSE FALSE
[49] FALSE TRUE TRUE FALSE FALSE FALSE FALSE TRUE FALSE FALSE TRUE FALSE
[61] TRUE TRUE FALSE TRUE FALSE FALSE TRUE TRUE TRUE TRUE FALSE FALSE
[73] TRUE TRUE TRUE FALSE FALSE FALSE TRUE FALSE TRUE TRUE TRUE FALSE
[85] FALSE

```

```

fruit.candy <- candy[fruit.indxs,]
fruit.candy

```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Air Heads	0	1	0	0	0
Caramel Apple Pops	0	1	1	0	0
Chewey Lemonhead Fruit Mix	0	1	0	0	0
Chiclets	0	1	0	0	0
Dots	0	1	0	0	0
Dum Dums	0	1	0	0	0
Fruit Chews	0	1	0	0	0
Fun Dip	0	1	0	0	0
Gobstopper	0	1	0	0	0
Haribo Gold Bears	0	1	0	0	0
Haribo Sour Bears	0	1	0	0	0
Haribo Twin Snakes	0	1	0	0	0
Jawbusters	0	1	0	0	0
Laffy Taffy	0	1	0	0	0
Lemonhead	0	1	0	0	0
Lifesavers big ring gummies	0	1	0	0	0
Mike & Ike	0	1	0	0	0
Nerds	0	1	0	0	0
Nik L Nip	0	1	0	0	0
Now & Later	0	1	0	0	0
Pop Rocks	0	1	0	0	0
Red vines	0	1	0	0	0
Ring pop	0	1	0	0	0
Runts	0	1	0	0	0
Skittles original	0	1	0	0	0
Skittles wildberry	0	1	0	0	0
Smarties candy	0	1	0	0	0
Sour Patch Kids	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0

Starburst	0	1	0	0	0
Strawberry bon bons	0	1	0	0	0
Super Bubble	0	1	0	0	0
Swedish Fish	0	1	0	0	0
Tootsie Pop	1	1	0	0	0
Trolli Sour Bites	0	1	0	0	0
Twizzlers	0	1	0	0	0
Warheads	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
	crisped	rice	wafer	hard bar	pluribus sugarpercent
Air Heads		0	0	0	0.906
Caramel Apple Pops		0	0	0	0.604
Chewey Lemonhead Fruit Mix		0	0	0	1.0732
Chiclets		0	0	0	1.046
Dots		0	0	0	1.0732
Dum Dums		0	1	0	0.732
Fruit Chews		0	0	0	1.127
Fun Dip		0	1	0	0.732
Gobstopper		0	1	0	1.0906
Haribo Gold Bears		0	0	0	1.0465
Haribo Sour Bears		0	0	0	1.0465
Haribo Twin Snakes		0	0	0	1.0465
Jawbusters		0	1	0	1.093
Laffy Taffy		0	0	0	0.220
Lemonhead		0	1	0	0.046
Lifesavers big ring gummies		0	0	0	0.267
Mike & Ike		0	0	0	1.0872
Nerds		0	1	0	1.0848
Nik L Nip		0	0	0	1.0197
Now & Later		0	0	0	1.0220
Pop Rocks		0	1	0	1.0604
Red vines		0	0	0	1.0581
Ring pop		0	1	0	0.732
Runts		0	1	0	1.0872
Skittles original		0	0	0	1.0941
Skittles wildberry		0	0	0	1.0941
Smarties candy		0	1	0	1.0267
Sour Patch Kids		0	0	0	1.0069
Sour Patch Tricksters		0	0	0	1.0069
Starburst		0	0	0	1.0151
Strawberry bon bons		0	1	0	1.0569
Super Bubble		0	0	0	0.162
Swedish Fish		0	0	0	1.0604

Tootsie Pop	0	1	0	0	0.604
Trolli Sour Bites	0	0	0	1	0.313
Twizzlers	0	0	0	0	0.220
Warheads	0	1	0	0	0.093
Welch's Fruit Snacks	0	0	0	1	0.313

	pricepercent	winpercent
Air Heads	0.511	52.34146
Caramel Apple Pops	0.325	34.51768
Chewey Lemonhead Fruit Mix	0.511	36.01763
Chiclets	0.325	24.52499
Dots	0.511	42.27208
Dum Dums	0.034	39.46056
Fruit Chews	0.034	43.08892
Fun Dip	0.325	39.18550
Gobstopper	0.453	46.78335
Haribo Gold Bears	0.465	57.11974
Haribo Sour Bears	0.465	51.41243
Haribo Twin Snakes	0.465	42.17877
Jawbusters	0.511	28.12744
Laffy Taffy	0.116	41.38956
Lemonhead	0.104	39.14106
Lifesavers big ring gummies	0.279	52.91139
Mike & Ike	0.325	46.41172
Nerds	0.325	55.35405
Nik L Nip	0.976	22.44534
Now & Later	0.325	39.44680
Pop Rocks	0.837	41.26551
Red vines	0.116	37.34852
Ring pop	0.965	35.29076
Runts	0.279	42.84914
Skittles original	0.220	63.08514
Skittles wildberry	0.220	55.10370
Smarties candy	0.116	45.99583
Sour Patch Kids	0.116	59.86400
Sour Patch Tricksters	0.116	52.82595
Starburst	0.220	67.03763
Strawberry bon bons	0.058	34.57899
Super Bubble	0.116	27.30386
Swedish Fish	0.755	54.86111
Tootsie Pop	0.325	48.98265
Trolli Sour Bites	0.255	47.17323
Twizzlers	0.116	45.46628
Warheads	0.116	39.01190

Welch's Fruit Snacks 0.313 44.37552

```
fruit.win <- fruit.candy$winpercent  
fruit.win
```

```
[1] 52.34146 34.51768 36.01763 24.52499 42.27208 39.46056 43.08892 39.18550  
[9] 46.78335 57.11974 51.41243 42.17877 28.12744 41.38956 39.14106 52.91139  
[17] 46.41172 55.35405 22.44534 39.44680 41.26551 37.34852 35.29076 42.84914  
[25] 63.08514 55.10370 45.99583 59.86400 52.82595 67.03763 34.57899 27.30386  
[33] 54.86111 48.98265 47.17323 45.46628 39.01190 44.37552
```

```
fruit.mean <- mean(fruit.win)  
fruit.mean
```

```
[1] 44.11974
```

```
# 7. Which mean value is higher
```

```
choc.mean > fruit.mean
```

```
[1] TRUE
```

The mean chocolate candy rankings is higher than the ranking of fruity candy.

Q12. Is this difference statistically significant?

```
t.test(choc.win, fruit.win)
```

Welch Two Sample t-test

```
data:  choc.win and fruit.win  
t = 6.2582, df = 68.882, p-value = 2.871e-08  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 11.44563 22.15795  
sample estimates:  
mean of x mean of y  
 60.92153  44.11974
```

Based on two sample t-test, the difference between chocolate and fruity candy is statistically significant.

Q13. What are the five least liked candy types in this set?

```
candy %>% arrange(winpercent) %>% head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0

	crispedricewafer	hard bar	pluribus	sugarpercent	pricepercent	
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The five least liked candies are Nik L Nip, Boston Baked Beans, Chiclets, Super Bubble, and Jawbusters. I prefer using piping to answer this question because I find it easier to read the code.

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy %>% arrange(desc(winpercent)) %>% head(5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's Peanut Butter cup	1	0	0	1	0
Reese's Miniatures	1	0	0	1	0
Twix	1	0	1	0	0
Kit Kat	1	0	0	0	0
Snickers	1	0	1	1	1

	crispedricewafer	hard bar	pluribus	sugarpercent
--	------------------	----------	----------	--------------

Reese's Peanut Butter cup	0	0	0	0	0.720
Reese's Miniatures	0	0	0	0	0.034
Twix	1	0	1	0	0.546
Kit Kat	1	0	1	0	0.313
Snickers	0	0	1	0	0.546

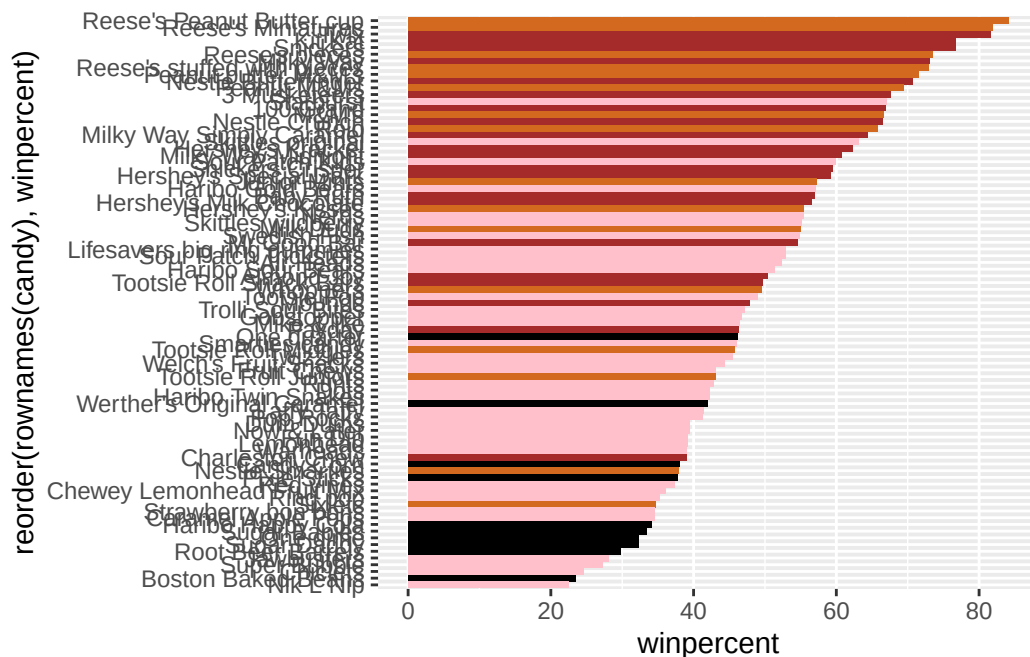
	pricepercent	winpercent
Reese's Peanut Butter cup	0.651	84.18029
Reese's Miniatures	0.279	81.86626
Twix	0.906	81.64291
Kit Kat	0.511	76.76860
Snickers	0.651	76.67378

The top 5 candies are Reese's Peanut Butter cup, Reese's Miniatures, Twix, Kit Kat, and Snickers.

Q15. Make a first barplot of candy ranking based on winpercent values.

```
my_cols <- rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] <- "chocolate"
my_cols[as.logical(candy$bar)] <- "brown"
my_cols[as.logical(candy$fruity)] <- "pink"
```

```
ggplot(candy) +aes(x = winpercent, y = reorder(rownames(candy), winpercent)) +geom_col(fill = my_cols)
```

Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy, based on the bar graph, is Sixlets.

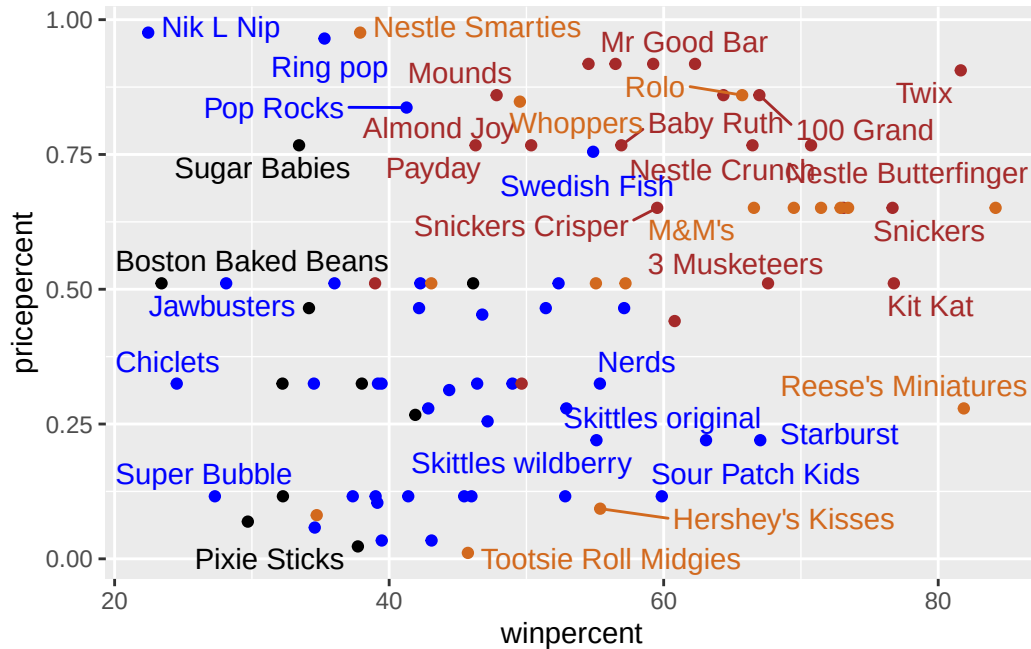
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

Winpercent vs Pricepercent

```
my_cols[as.logical(candy$fruity)] <- "blue"
ggplot(candy) +aes(winpercent, pricepercent, label = rownames(candy)) +geom_point(col = my_c
```

Warning: ggrepel: 50 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Fruity candy tends to be the highest rank for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
candy %>% arrange(desc(pricepercent)) %>% head(5) %>% arrange(winpercent)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Ring pop	0	1	0		0	0
Nestle Smarties	1	0	0		0	0
Hershey's Milk Chocolate	1	0	0		0	0
Hershey's Krackel	1	0	0		0	0
	crisped	rice	wafer	hard bar	pluribus	sugar
Nik L Nip		0	0	0	1	0.197
Ring pop		0	1	0	0	0.732
Nestle Smarties		0	0	0	1	0.267
Hershey's Milk Chocolate		0	0	1	0	0.430
Hershey's Krackel		1	0	1	0	0.430
	price	percent win	percent			

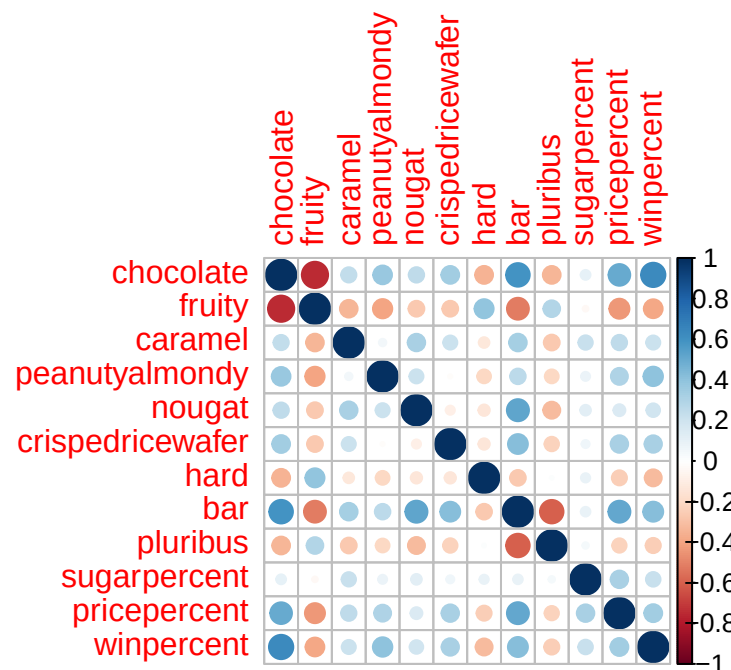
Nik L Nip	0.976	22.44534
Ring pop	0.965	35.29076
Nestle Smarties	0.976	37.88719
Hershey's Milk Chocolate	0.918	56.49050
Hershey's Krackel	0.918	62.28448

The top five most expensive candy types in the dataset are Nik L Nip, Nestle Smarties, Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate. The least popular among these five is Nik L Nip.

Correlation

```
cij <- cor(candy)
```

```
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruit and chocolate tend to be anti-correlated.

Q23. Similarly, what two variables are most positively correlated?

Chocolate and bar as well as chocolate and winpercent to be positively correlated.

Principal Component Analysis

The main function in base R for this is `prcomp()` and we want to set `scale=TRUE` here:

```
pca <- prcomp(candy, scale = T)
summary(pca)
```

Importance of components:

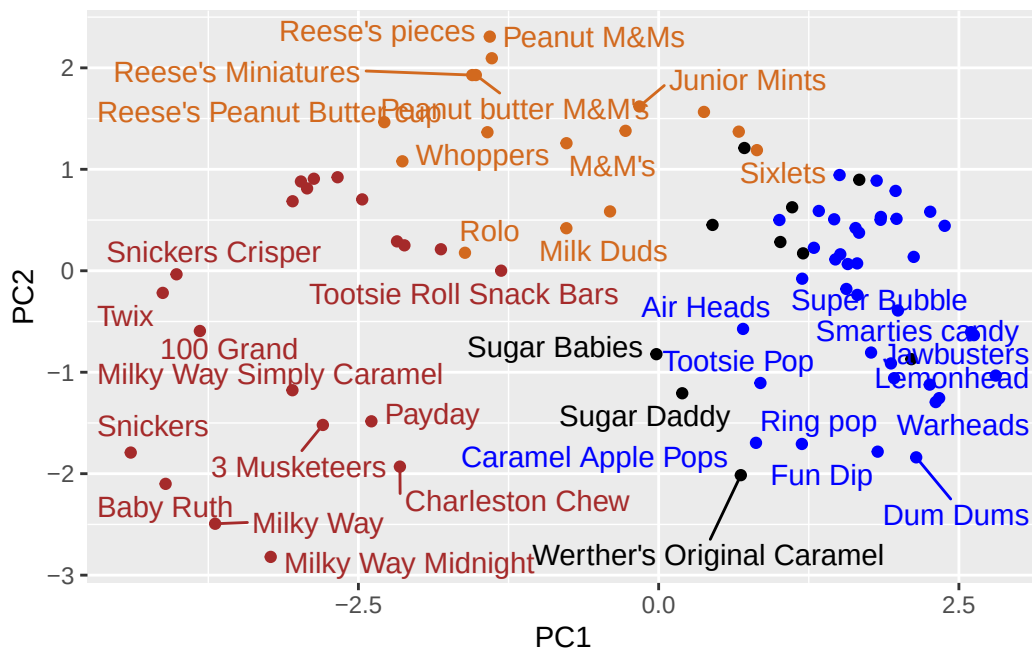
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

Let's look at our first main result figure - the "PC plot" on PC1 vs PC2

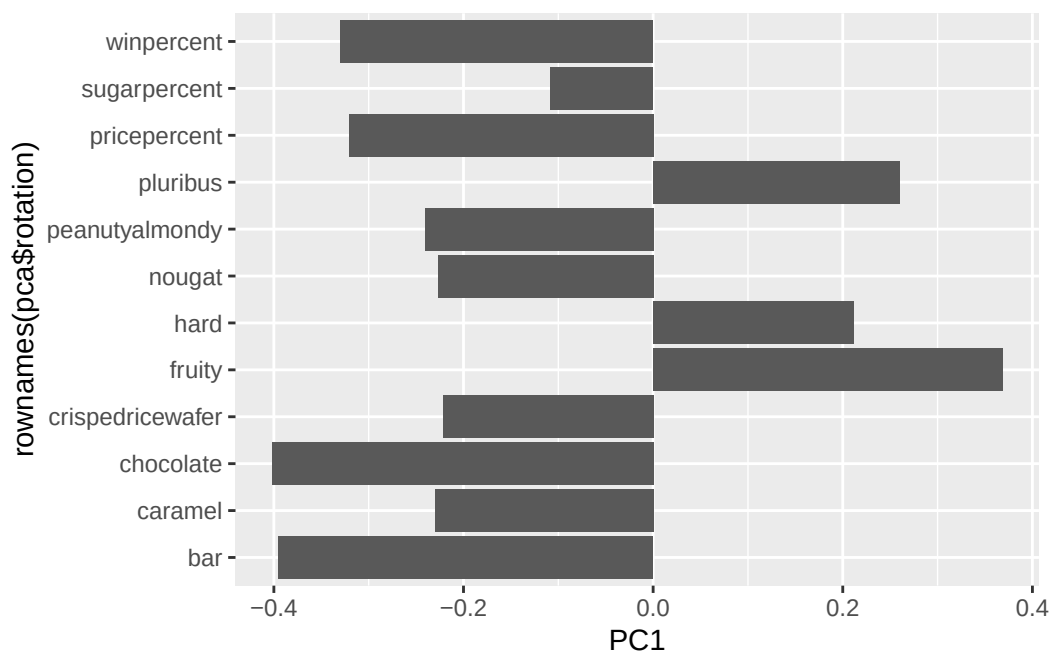
```
ggplot(pca$x) +aes(PC1, PC2, label = rownames(pca$x)) +geom_point(col = my_cols) +geom_text_1
```

Warning: ggrepel: 48 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Don't forget about your variable "loadings" - how the original variables contribute to your new PCs

```
ggplot(pca$rotation) + aes(PC1, rownames(pca$rotation)) + geom_col()
```



Q24. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you?

Pluribus, hardest, and fruity-ness are strongly picked up by PC1 in a positive direction. This does make sense since most fruity candies tend to be hard candies.